The University of Manchester

EART11200 - The Natural Scientist's Toolkit
Mathematics and Physics

Week 4 Practical - Optimisation and Properties of Derivatives (Solutions)

Course layout

Date	Week	Topic
27/01/2025	20	Calculating Gradients
03/02/2025	21	Introduction to Differentiation
10/02/2025	22	Differentiating Combinations of Functions
17/02/2025	23	Optimisation & Properties of Derivatives
24/02/2025	24	Second Order Derivatives & Root Finding
03/03/2025	25	Introduction to Integration
10/03/2025	26	Integration using Newtons Laws
17/03/2025	27	Integration by Substitution
24/03/2025	28	Integrating Second Order ODEs
31/03/2025	29	Vectors and Partial Differentiation
07/04/2025	30	Non-teaching week
14/04/2025	31	Easter break
21/04/2025	32	Easter break
28/04/2025	33	Revision: Differentiation
05/05/2025	34	Revision: Integration
Exam Period		

Intended Learning Outcomes

(1) Describe the properties of derivatives in the analysis of functions.

(2) Analyse the behavior of functions by applying differentiation in theoretical and real-world applications.

(3) Solve differentiated functions to determine stationary values in theoretical and real-world applications.

(4) Apply differentiation techniques to optimise functions in real-world problems.

Exercise 1: Stationary Values

- Find the stationary values of the following:

$$f(x) = \frac{x^3}{3} + x^2 - 15x + 1$$

- Look at the properties of the derivative to check whether they are maxima, minima, or saddle points.

Solution 1: Stationary Values

- Differentiate the function
$$\frac{df}{dx} = x^2 + 2x - 15$$
- Set derivative equal to zero and find the stationary points
$$\frac{df}{dx} = 0 \implies x^2 + 2x - 15 = 0 \implies (x+5)(x-3) = 0$$
- We have two stationary values at $x = -5$ and $x = 3$.

Solution 1: Stationary Values

- Evaluate the derivative either side of the stationary values:

$$x = -6 \implies \frac{df}{dx}(-6) = (-6)^2 + 2 \times (-6) - 15 = 9$$
$$x = 0 \implies \frac{df}{dx}(0) = 0^2 + 2 \times 0 - 15 = -15$$
$$x = 4 \implies \frac{df}{dx}(4) = 4^2 + 2 \times 4 - 15 = 9$$

- We see that $f(x)$ is
  - increasing when $x < -5$
  - decreasing when $-5 < x < 3$
  - increasing when $x > 3$
- Therefore, $x = -5$ is a maximum and $x = 3$ is a minimum.

Exercise 2: Optimisation

- A warehouse wants to construct a box with length l, width w, and height h which has volume:

$$V = l \times w \times h$$

They only have 10m^2 of material to use in construction of the box so the surface area:

$$10 = 2lw + 2wh + 2lh$$

The base of the box must be rectangular, with the length of the box is the same as the width (i.e. $l = w$).

Exercise 2: Optimisation

- We rearrange the surface area equation ($10 = 2lw + 2wh + 2lh$ and $l = w$) to find an equation for h:

$$h = \frac{5 - w^2}{2w}$$

  (a) Substitute the constraint for h in the volume:

$$V = l \times w \times h$$

  (b) Find the optimum value of w that maximises the volume (i.e. the stationary value).

Solution 2: Optimisation

- Substituting h and $l = w$ into the volume equation gives us

$$V(w) = w^2 \left(\frac{5 - w^2}{2w} \right) = \frac{1}{2}(5w - w^3)$$

- To optimise the box dimensions we need to find the stationary values of the volume:

$$V(w) = \frac{1}{2}(5w - w^3) \qquad \frac{dV}{dw} = \frac{1}{2}(5 - 3w^2)$$

- Set derivative equal to zero and find the stationary points:

$$\frac{1}{2}(5 - 3w^2) = 0 \implies 3w^2 = 5 \implies w = \sqrt{\frac{5}{3}} \approx 1.29m$$

Solution 2: Optimisation

- The volume of the box is optimised with

$$l = w = \sqrt{\frac{5}{3}}m \qquad h = \frac{5 - \frac{5}{5}}{2\frac{5}{3}} = \sqrt{\frac{5}{3}}m$$

- The optimum box is a cube!
- The optimum volume is

$$V(w) = \frac{1}{2}\left(5\sqrt{\frac{5}{3}} - \left(\sqrt{\frac{5}{3}}\right)^3\right) = 2.15m^3$$

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